

The Choptuik Problem from the Classical Einstein Equations: Direct Derivation from the Hilbert Action and Numerical Solution

einstein_direct module of the choptuik_ac_bc project

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Abstract

The Choptuik problem (1993) — critical gravitational collapse of a massless scalar field in spherical symmetry: near the black-hole threshold the mass scales as $M_{\text{BH}} \propto (p - p_*)^\gamma$ with the universal exponent $\gamma \approx 0.374$, and the critical solution is discretely self-similar (DSS) with echoing period $\Delta \approx 0.737$. This work solves the problem “directly”: the field equations are derived from the Hilbert action $S = (2\kappa)^{-1} \int \sqrt{-g} R d^4x + S_m$ with the matter stress–energy tensor in Hilbert form $T_{\mu\nu} = -(2/\sqrt{-g}) \delta S_m / \delta g^{\mu\nu}$, reduced to a 1+1 double-null characteristic system that is *machine-verified* (SymPy) against the exact Roberts–Oshiro solution with residuals $\sim 10^{-41}$, and integrated by a second-order solver (validation: flat space to 10^{-14} ; Roberts solution — convergence order $p = 2.0$ – 2.1). The critical amplitude $A_* = 0.080533$ is located; the exponent measured on a fixed grid, $\gamma_{\text{num}} = 0.11 \pm 0.11$, exhibits a resolution floor; our analysis shows that a percent-level verification of γ (and a fortiori testing the hypothesis $b_{\text{Ch}} = 1 - \cos(2\pi/7) = 0.3765$ against $\gamma = 0.374$) requires production-grade regridding at the Choptuik level. All code, derivations and data are included in the repository.

1 The Hilbert derivation

Starting from the Hilbert action $S = (2\kappa)^{-1} \int \sqrt{-g} R d^4x - \frac{1}{2} \int \sqrt{-g} (\nabla\Phi)^2 d^4x$ with $\kappa = 8\pi G$, variation w.r.t. $g^{\mu\nu}$ gives $G_{\mu\nu} = \kappa T_{\mu\nu}$ with the *Hilbert stress–energy tensor* $T_{\mu\nu} = \partial_\mu\Phi \partial_\nu\Phi - \frac{1}{2}g_{\mu\nu}(\nabla\Phi)^2$, and variation w.r.t. Φ gives $\square_g\Phi = 0$. The Hilbert identity $\nabla_\mu T^\mu{}_\nu \equiv 0 \Leftrightarrow \square_g\Phi = 0$ is verified symbolically.

For the spherically symmetric metric in double-null coordinates $ds^2 = -\alpha(u, v)^2 du dv + r(u, v)^2 d\Omega^2$ with $\omega = \ln \alpha$, the projections form a closed system (all machine-verified):

$$\begin{aligned} \text{(SC)} \quad \Phi_{,uv} &= -\frac{r_{,u}\Phi_{,v} + r_{,v}\Phi_{,u}}{r}, \\ \text{(UV)} \quad r_{,uv} &= -\frac{r_{,u}r_{,v} + \alpha^2/4}{r} = -\frac{\alpha^2 m}{2r^2}, \\ \text{(C1,C2)} \quad r_{,uu} &= 2\omega_{,u}r_{,u} - \frac{\kappa}{2}r\Phi_{,u}^2, \quad r_{,vv} = 2\omega_{,v}r_{,v} - \frac{\kappa}{2}r\Phi_{,v}^2, \\ \text{(TH)} \quad \omega_{,uv} &= \frac{\alpha^2 m}{2r^3} - \frac{\kappa}{2}\Phi_{,u}\Phi_{,v}, \end{aligned}$$

with the Misner–Sharp mass $1 - 2m/r = -4r_{,u}r_{,v}/\alpha^2$ and the regularizing identities $(r^2)_{,uv} = -\alpha^2/2$, $m_u = -\kappa r^2\Phi_u^2 r_v/\alpha^2$, $m_v = -\kappa r^2\Phi_v^2 r_u/\alpha^2$.

Machine verification. SymPy builds the metric, all Christoffel symbols, the Ricci and Einstein tensors and the Hilbert tensor, projects the field equations and solves for the highest derivatives; the result coincides exactly with the system above. The exact Roberts–Oshiro solution (corrected form, Burko gr-qc/9608061) $r^2 = \frac{1}{4}[(1 - 4\sigma^2)v^2 - 2uv + u^2]$, $\Phi = \frac{1}{2} \ln |(1 - 2\sigma - u/v)/(1 + 2\sigma - u/v)|$, $\alpha = 1$ is substituted: all five equations are satisfied with the maximum residual $4 \cdot 10^{-41}$ (mpmath, 50 digits).

2 Numerical method and validation

A Goursat-characteristic march in v (Heun/RK2 with an iterated corrector; the pair (Φ, t) updated via an implicit linear ODE solve along u ; exact center regularity $t = s, p = -q$ at $r = 0$; the Misner–Sharp mass evolved by the regularizing identities). Validation: flat space preserved to $8.8 \cdot 10^{-15}$; the Roberts–Oshiro solution evolved within a singularity-free strip gives clean second-order convergence ($p = 2.03$ – 2.09 for r , 1.99 for Φ); the C1 constraint violation decreases quadratically.

3 Results

For the Choptuik family $\Phi(u_0, v) = A \exp[-(v - v_p)^2/2\sigma^2]$ ($u_0 = -1, v_p = 0.5, \sigma = 0.1$) the critical amplitude, bisected on the singularity criterion, is $A_* = 0.0805333$ (grid 1600^2 ; biased by the horizon-nucleation threshold at $\delta \sim 10^{-4}$). Sub- and supercritical evolutions show the expected qualitative picture (dispersal vs. trapped surface $2m/r \geq 1$).

Honest comparison with the framework constant

The exponent measured on the fixed grid, $\gamma_{\text{num}} = 0.11 \pm 0.11$, is floored by the horizon-nucleation threshold $M \geq 4 du$ and by critical slowing down: near-critical echoes shrink below the grid scale long before the apparent-horizon mass saturates. Resolving the universal $\gamma = 0.374$ (and testing $b_{\text{Ch}} = 1 - \cos(2\pi/7) = 0.376510$ against it at the literature gap of $+0.67\%$) requires a Choptuik-grade regridding campaign (3–4 decades in $A - A_*$). A regridding prototype (`zoom_solver.py`) is included. What this module does establish: (a) the Hilbert-derived system is correct and machine-checked; (b) the second-order solver is validated on exact solutions; (c) the critical point and the qualitative critical behaviour are reproduced.

4 Reproducibility

`sympy_derivation.py` (symbolic derivation + Roberts check), `solver.py` (flat-space self-test), `roberts_test.py` (second-order convergence), `choptuik_scaling.py` (bisection + scaling fit), `zoom_solver.py` (regridding prototype), `figures.py` (RU/EN figures). Units: $4\pi G = 1, \kappa = 2$ (Burko normalization). One 1200^2 run takes ~ 1 s (NumPy-vectorized).

References

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